CS 6362 Advanced Machine Learning, Fall 2026
Assignment 2: β-VAE: Learning Basic Visual Concepts with a Constrained Variational Framework

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1 Problem Statement and Methodology

1.1 Problem Statement

We observe images $\mathbf{x} \in \mathbb{R}^N$ that are generated from a set of ground-truth factors. These are split into conditionally independent factors $\mathbf{v} \in \mathbb{R}^K$, with $\log p(\mathbf{v}|\mathbf{x}) = \sum_k \log p(v_k|\mathbf{x})$, and other factors $\mathbf{w} \in \mathbb{R}^H$, so that $p(\mathbf{x}|\mathbf{v}, \mathbf{w}) = \mathrm{Sim}(\mathbf{v}, \mathbf{w})$. For dSprites the factors are shape, scale, rotation, and x and y position.

The goal is to learn, from $\mathbf{x}$ alone, a latent code $\mathbf{z} \in \mathbb{R}^M$ ($M \geq K$) that can generate the data. We want this code to be *disentangled*: each latent z_j should respond to one factor v_k and stay roughly unchanged when the other factors change. We learn an encoder $q_\phi(\mathbf{z}|\mathbf{x})$ and a decoder $p_\theta(\mathbf{x}|\mathbf{z})$, with prior $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, I)$.

1.2 Model and Methodology

The model is a standard VAE with one change: the KL term gets a weight β. They start from a constrained problem. Maximize reconstruction while keeping the posterior close to the prior:

$$\max_{\phi,\theta} \; \mathbb{E}_{\mathbf{x}\sim\mathcal{D}}\Big[\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\big[\log p_\theta(\mathbf{x}|\mathbf{z})\big]\Big] \quad \text{s.t.} \quad D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, p(\mathbf{z})) < \epsilon. \tag{1}$$

Writing the Lagrangian and dropping the constant $\beta\epsilon$ gives the β-VAE objective

$$\mathcal{L}(\theta, \phi; \mathbf{x}, \mathbf{z}, \beta) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\big[\log p_\theta(\mathbf{x}|\mathbf{z})\big] - \beta\, D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, p(\mathbf{z})). \tag{2}$$

$\beta = 1$ is the usual VAE (the ELBO). The paper claims that $\beta > 1$ gives more disentangled latents, at the cost of worse reconstructions. Since there is no principled way to pick β, they also propose a supervised disentanglement metric for choosing it.

They first show qualitative latent traversals on CelebA, 3D chairs and 3D faces, comparing against VAE, InfoGAN and DC-IGN. For quantitative results they use dSprites, where the true factors are known. That is the setting of Figure 7, which compares $\beta = 4$ with $\beta = 1$.

1.3 Mathematical Derivations and Intuition

With a diagonal Gaussian encoder $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \mathrm{diag}(\boldsymbol{\sigma}^2))$, the KL term has the closed form

$$D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, \mathcal{N}(\mathbf{0}, I)) = \frac{1}{2}\sum_{j=1}^{M}\Big(\mu_j^2 + \sigma_j^2 - \log\sigma_j^2 - 1\Big). \tag{3}$$

Each latent pays a separate cost. A latent that carries no information can set $\mu_j = 0$, $\sigma_j = 1$ and pay nothing. A latent that is used has to move away from that, and the cost grows with how precisely it encodes x.

Here is how I think about why a larger β helps. The KL term acts as a budget on how much information $\mathbf{z}$ can carry about $\mathbf{x}$. With a tight budget, the model only keeps what reduces the reconstruction error the most, and it wants to spend as little as possible to do it. Because the prior is isotropic and factorized, the cheapest way to do this is to put each independent factor of variation into its own latent and switch off the rest. That is a disentangled code. With $\beta = 1$ the budget is looser, so the model is free to spread factors across many latents.

The price is reconstruction quality. A tighter channel throws away fine detail, so the reconstructions get blurrier as β grows. This is also why the log-likelihood is not a good way to compare models here.

The disentanglement metric works like this. Fix one factor y and sample pairs of images that share that factor but differ in the others. Encode both images and average $|\mathbf{z}_1 - \mathbf{z}_2|$ over L pairs. A simple linear classifier then has to predict which factor was fixed. If the code is disentangled, the latent for the fixed factor will have a near-zero difference, and the classifier can tell easily.

The authors also report a normalized $\beta_{\text{norm}} = \beta M/N$ so that values are comparable across input and latent sizes.

2 Reproduction of Figure 7 (A, B)

2.1 Original Figure

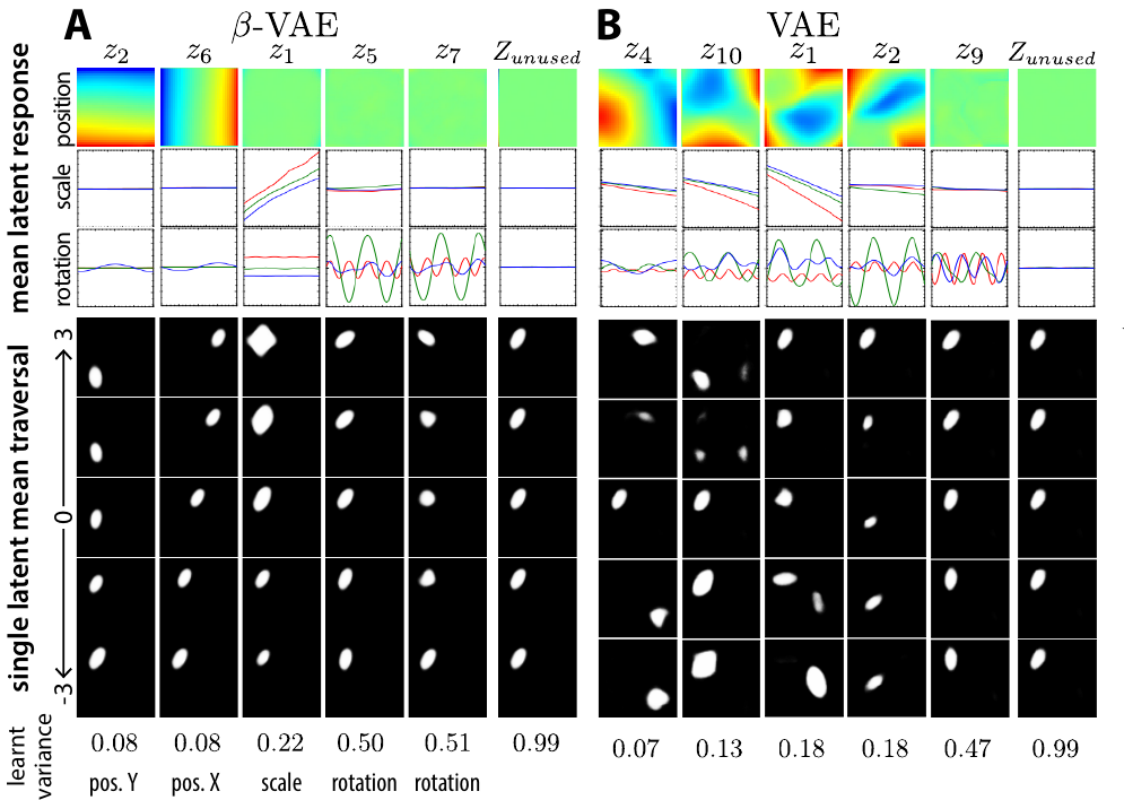


Figure 1: Figure 7A (β-VAE, $\beta = 4$) and 7B (VAE, $\beta = 1$) of Higgins et al. (2017). Each column is one latent, ordered by its learnt Gaussian variance (bottom numbers). Rows 1–3 show the mean latent response against position, scale and rotation (square red, oval green, heart blue). Rows 4–8 show decoded images as one latent is moved from $+3$ to -3 while the other nine stay at the values inferred for one image.

2.2 LLM Details and Code

- **LLM (name and version):** Claude Opus 5.5 (claude.ai chat interface)
- **Shared conversation:** https://claude.ai/share/0b99bf62-4b32-41fd-a426-9d683489f0e7
- **GitHub repository:** https://github.com/souvik7701/ML-Assignment-2. The notebook `Paper_3.ipynb` reproduces the figure.

During Phases 1–3 the LLM was not given any existing code. In its first answer it stated that it did not look at any existing implementation, and that the only outside resource it used was DeepMind's public dSprites release, which is the dataset the paper describes. I ran all the training myself on a Colab GPU.

2.3 Phase 1: Prompting the LLM with the Paper

I gave the LLM the paper and asked it to reproduce Figures 7A and 7B "following the rubric outlined in Section 1.1", split into Colab cells. I asked it to mount my Drive, download the data there, and cache all results so that a disconnect would not lose finished work. I told it not to use or look at any existing implementation. Section 1.1 is the assignment's rubric, which I had not pasted, and the LLM pointed out that the paper itself has no Section 1.1. It worked from what the paper gives for Figure 7: Section 4.2, the "2D shapes (VAE)" row of Table 1, and the figure caption.

The notebook it wrote follows the paper closely:

- **Model and training:** the fully connected encoder and decoder, 10 Gaussian latents, Bernoulli output and Adagrad optimizer from Table 1, with the KL term weighted by $\beta = 4$ (panel A) or $\beta = 1$ (panel B).
- **Figure rows (caption):** each latent's mean response against position, scale and rotation, then the traversal rows, then the learnt variance.

It also made its own choices where the paper is silent, such as the batch size, training length, the seed image for the traversals, and how to choose and order the columns (listed under Implementation Choices). It added automatic factor labels under panel A; the paper labeled its columns by hand.

2.4 Phase 2: Reviewing the LLM Output

The overall result looked close to the paper, but Figure 7A was only partly reproduced. Position and rotation came out as separate latents, but there was no clean scale latent. The latent that responds most to scale (z_3) also responds to position. I saw two possible problems:

- **The selection rule.** Ranking latents by mean σ^2 ignores the posterior means, so it may not identify the informative and unused latents correctly. The paper itself identifies them by KL (Sec. 4.2): "the most informative latent units z_m of β-VAE have the highest KL divergence from the unit Gaussian prior, while the uninformative latents have KL divergence close to zero."
- **Training.** The $\beta = 4$ run had not fully converged. Near the end, the reconstruction error was still falling and the KL still rising.

I asked the LLM to select latents by mean per-latent KL while keeping the caption's ordering by learnt variance, to check whether any minimal change to training or plotting could help, and, if scale still did not separate, to explain why rather than force a latent to represent it.

2.5 Phase 3: Good-Faith Reproduction

The second version (v2) keeps the trained models and changes only how the columns are chosen. For every latent it computes the mean KL over the whole dataset,

$$\overline{\mathrm{KL}}_i = \frac{1}{2}\,\mathbb{E}_{\mathbf{x}}\big[\mu_i^2 + \sigma_i^2 - 1 - \log \sigma_i^2\big], \tag{4}$$

shows the five latents with the highest KL sorted by ascending σ^2 as in the caption, and puts the lowest-KL latent last as z_{unused}. As a sanity check, the per-latent KLs add up to the final KL in the training logs.

For the $\beta = 4$ model the KL rule picks the same columns as the variance rule (Table 1). Five latents carry information and five have KL ≈ 0, so the two rules agree. The selection rule was therefore not what kept scale from separating: z_3 is genuinely entangled. Its variance is split roughly equally between scale, position X and position Y, its position map is a radial bowl, and its traversal changes both the size and the location of the object.

				fraction of variance of μ_i explained by				
model	latent	KL	σ^2	shape	scale	rotation	pos. X	pos. Y
$\beta = 4$	z_6	3.29	0.002	0.00	0.00	0.00	0.01	0.92
	z_7	3.26	0.002	0.00	0.00	0.00	0.88	0.01
	z_3	2.49	0.008	0.06	0.28	0.00	0.29	0.21
	z_{10}	1.25	0.117	0.00	0.00	0.57	0.00	0.00
	z_2	1.22	0.126	0.00	0.00	0.10	0.00	0.00
	5 others	0.00	0.99	–	–	–	–	–
$\beta = 1$	z_6	4.05	0.000	0.00	0.00	0.01	0.47	0.40
	z_7	3.94	0.001	0.00	0.01	0.00	0.50	0.36
	z_3	3.43	0.001	0.04	0.11	0.00	0.28	0.46
	z_1	2.47	0.011	0.01	0.01	0.14	0.06	0.06
	z_2	2.36	0.011	0.02	0.03	0.09	0.02	0.03
	z_9	2.26	0.014	0.00	0.00	0.45	0.00	0.01
	4 others	≤ 0.002	≥ 0.95	–	–	–	–	–

Table 1: Per-latent statistics of the final models. The variance fractions come from the LLM's labeling heuristic, not from the paper. They miss variance that comes from interactions between factors. For z_2 of $\beta = 4$ they add up to only 0.10, which is why it is labeled "mixed" even though its plots show a rotation latent (Figure 2).

The LLM found nothing in the training code that departs from Table 1. It suggested two changes to settings the paper does not specify: training the $\beta = 4$ model longer (to 1M steps) and training more seeds, since the paper trained 10 replicas per setting. In a third answer it put two extra $\beta = 4$ seeds and the longer run into the notebook. I decided not to run this version. The v2 results already show the comparison the figure is meant to show, and an extra seed with a clean scale latent would not change that conclusion. I later added two short checks that use the trained models as they are (Section 2.8). They do not change the figure.

2.6 Phase 4: Comparison with an Existing Implementation

The authors did not release an implementation of β-VAE. DeepMind's dSprites repository (https://github.com/deepmind/dsprites-dataset) contains only the dataset and a notebook that loads it. Its README confirms the setup the code assumes: every combination of the factor values appears exactly once, stored in a fixed order. The notebook relies on that order to average over factors, and it checks the order when loading the data. I did not consult any third-party re-implementation, and I made no changes in this phase.

2.7 Reproduced Figure

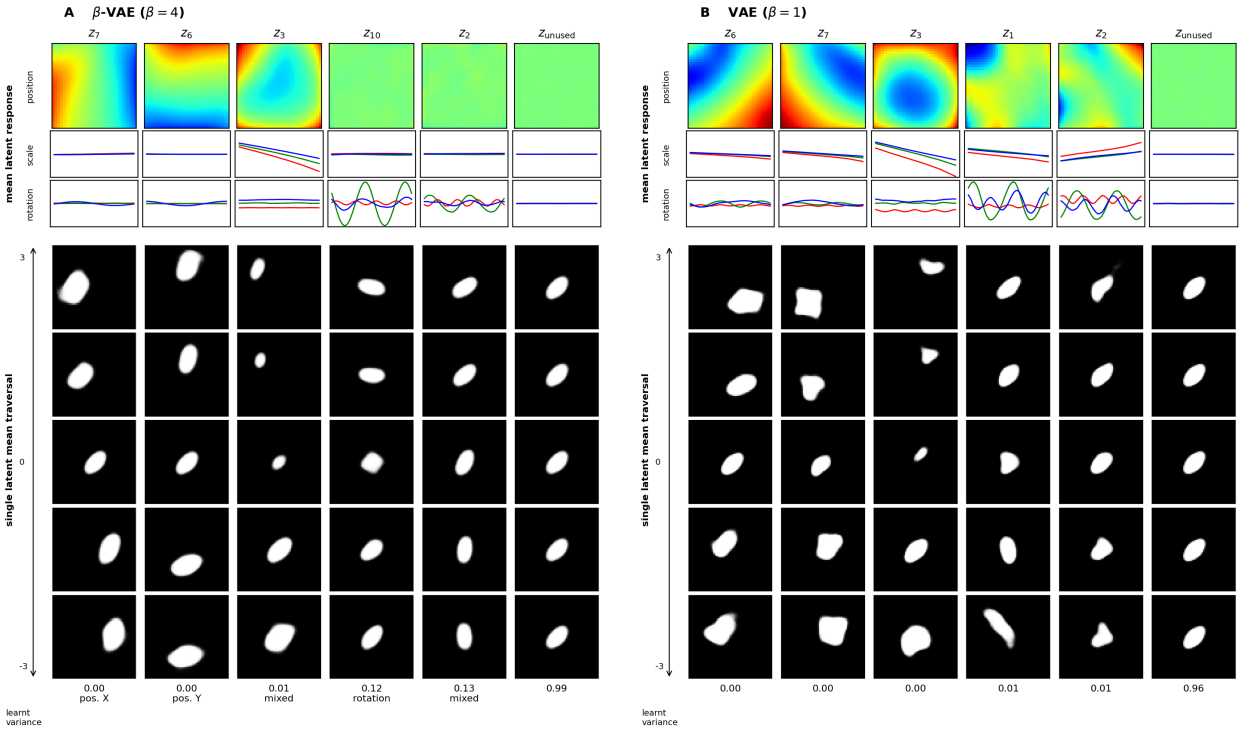


Figure 2: My reproduction (v2, one seed per β). Left: $\beta = 4$. Right: $\beta = 1$. The labels under panel A come from the automatic heuristic. Latent indices are arbitrary and do not correspond to the paper's.

Panel A ($\beta = 4$). Table 2 compares the columns. The position latents came out well. z_7 and z_6 are clean x and y latents, like z_6 and z_2 in the paper: their position maps are smooth gradients along one axis, their scale and rotation rows are flat, and their traversals move the object sideways or up and down (with some growth or deformation at the ± 3 ends). The rotation latents z_{10} and z_2 are also a good match. They respond only to rotation, and their oscillations follow each shape's symmetry, as the paper describes: one cycle for the heart, two for the oval and four for the square. The two curves are shifted against each other like the paper's cos/sin pair. The last column is flat in every row and its traversal leaves the image unchanged, as in the paper.

Scale is the main difference. In the paper z_1 is a clean scale latent, with monotone scale curves and a traversal that only changes the size. My z_3 has the monotone scale curves too, but also a radial bowl in its position map, and its traversal moves the object while it changes size.

paper (A)	pos. Y 0.08	pos. X 0.08	scale 0.22	rotation 0.50	rotation 0.51	unused 0.99
mine (A)	pos. X 0.00	pos. Y 0.00	scale+pos. 0.01	rotation 0.12	rotation 0.13	unused 0.99
paper (B)	mixed 0.07	mixed 0.13	mixed 0.18	mixed 0.18	rotation 0.47	unused 0.99
mine (B)	mixed 0.00	mixed 0.00	mixed 0.00	mixed 0.01	mixed 0.01	unused 0.96

Table 2: What each column encodes, with its learnt variance σ^2.

Panel B ($\beta = 1$). As in the paper, the VAE is entangled. z_6 and z_7 each mix x and y position and have diagonal position maps, z_3 mixes scale and position, and z_1 and z_2 respond to rotation but also to position and scale. Their traversals change several things at once, and the object often turns into a shape that is not in the dataset (square- or heart-like blobs from an oval). The paper notes one exception in its VAE, z_9, which learnt rotation. My VAE has a similar latent (z_9, rotation fraction 0.45), but it is not shown: this model has six informative latents, and the figure has room for five.

The learnt variances are much smaller than the paper's. My informative latents have σ^2 of 0.00–0.13, against 0.08–0.51 in the paper. So my encoder is far more certain, and each latent carries more information. A rough estimate from the KL formula puts the paper's informative latents at less than half of my model's total KL.

2.8 Failures, Diagnosis, and Corrections

The only real failure is that scale is not separated in panel A, and Phase 3 ruled out the selection rule as the cause. Three explanations remain. The first is luck: the paper trained 10 replicas per setting and discarded the bottom half by its metric (App. A.4), so Figure 7A is one of the better models from a pool, while I trained a single seed and cannot tell whether a mixed scale latent is typical. The second is the data. Scale is the most coarsely sampled factor, with 6 values against 32 or 40 for the others, and Appendix A.8 reports that lower sampling density reduces disentanglement. The third is training. The paper gives neither the batch size nor the training length, and my models carry much more information per latent than the paper's (see the variances above), which suggests a different training regime. The $\beta = 4$ run was also still slowly improving at 300k steps.

Diagnostic 1: how far the traversal goes. The LLM suggested that the ± 3 traversal might push the decoder beyond the values the encoder actually produces, which would explain why the position traversals also change size at the ends. To check this, I measured the spread of each shown latent's μ_i over the whole dataset. For every informative latent of both models, almost all values lie within about ± 2.5, so the ± 3 rows show codes that very few images produce. Such codes are rare but not impossible, which fits the small distortions at the ends of the traversals. The unused latents stay close to zero for every image, and the decoder ignores them.

Diagnostic 2: the paper's disentanglement metric. I also computed the metric of Section 3 for both models. For each sample it picks one of the four factors used in App. A.4 (scale, rotation, x, y), draws L pairs of images that share that factor and differ randomly in the others, and averages $|\mu(\mathbf{x}_1) - \mu(\mathbf{x}_2)|$ over the pairs. A linear classifier then predicts which factor was fixed, and I report its accuracy on held-out vectors. The paper does not give L or the size of the training set, so these are my choices (listed under Implementation Choices). Training the classifier longer barely changed the scores.

	β-VAE ($\beta = 4$)	VAE ($\beta = 1$)
mine (one model, 3 classifier seeds)	$94.52 \pm 0.07\%$	$74.99 \pm 0.49\%$
paper, Fig. 6 (top half of 10 replicas $\times$ 3 classifier seeds)	$99.23 \pm 0.1\%$	$61.58 \pm 0.5\%$

Table 3: Disentanglement metric score. My $\pm$ only covers the classifier seed, not the training seed.

The direction is clearly reproduced: $\beta = 4$ scores about 20 points higher than $\beta = 1$. The gap is smaller than the paper's 38 points. My $\beta = 4$ model scores lower than the paper's. This probably comes from its mixed scale latent: when scale is fixed, z_3 still varies with position, so that case is harder to recognize. My $\beta = 1$ model scores higher than the paper's VAE. With one model per β and none of the paper's selection of the best replicas, the exact numbers mean less than the direction.

2.9 Required Questions

What happens to $\mathcal{L}_\beta$ for $\beta > 1$? For $\beta < 1$? Write the objective as the usual ELBO plus a correction:

$$\mathcal{L}_\beta = \underbrace{\mathbb{E}_q[\log p_\theta(\mathbf{x}|\mathbf{z})] - D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, p(\mathbf{z}))}_{\text{ELBO} \le \log p_\theta(\mathbf{x})} - (\beta - 1)\, D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, p(\mathbf{z})). \tag{5}$$

Since the KL is never negative:

- $\beta > 1$**:** $\mathcal{L}_\beta \le \text{ELBO} \le \log p_\theta(\mathbf{x})$. It is still a lower bound on the log-likelihood, but a looser one. Every nat of information that $\mathbf{z}$ carries about $\mathbf{x}$ now costs β instead of 1, so the model keeps only the information that reduces reconstruction error the most. Seen as the Lagrangian of the constrained problem, this is a tighter capacity constraint. It pushes the posterior toward the factorized prior, switches off latents that are not needed, and makes reconstructions blurrier. As $\beta \to \infty$ the posterior collapses onto the prior and $\mathbf{z}$ carries no information.
- $\beta < 1$**:** $\mathcal{L}_\beta \ge \text{ELBO}$, so it is no longer guaranteed to be a lower bound on $\log p_\theta(\mathbf{x})$. Information is cheap, so the model favors reconstruction. As $\beta \to 0$ it becomes an autoencoder with noisy codes, with almost no pressure toward an independent, prior-like code, so disentanglement is not encouraged.

How are the latents in Figure 7 ordered, and how is each traversal generated? The caption says each column is a latent "ordered according to the learnt Gaussian variance (last row)", from the most certain (smallest σ^2, most informative) to the unused latent ($\sigma^2 \approx 1$). In my code the five highest-KL latents are sorted by ascending mean σ^2, and the lowest-KL latent comes last. For a traversal, one image (the seed image) is encoded to get its posterior mean $\boldsymbol{\mu}$. For column i, latent z_i is set to $+3, +1.5, 0, -1.5, -3$ (three prior standard deviations around the prior mean 0) while the other 9 latents stay at their values in $\boldsymbol{\mu}$. Each code is decoded, and the Bernoulli mean (the sigmoid of the decoder output) is shown as the image.

What evidence do Figures 7A and 7B give for the central claim, and does my reproduction show the same difference? In a disentangled code each latent should depend on one factor only. Figure 7 shows this in two ways. In the top rows, a disentangled latent has a response that changes along one factor and is flat along the others. In the traversals, moving a disentangled latent changes one property of the image and keeps the shape. In 7A almost every column passes both tests. In 7B most latents respond to two or more factors, and their traversals change position, size and shape together. Since the two models differ only in β, this supports the claim that the stronger KL pressure produces the more disentangled code. My reproduction shows the same difference for position and rotation, a weaker version for scale, and a clear gap in the paper's own metric (94.5% against 75.0%). It is still one model per setting, so it shows that the effect occurs, not how reliable it is.

Would increasing β further help, and what happens to reconstructions? Up to a point, a larger β should give cleaner factors, because each latent becomes more expensive and the model has more reason to use one latent per factor. Beyond that point it should hurt. The paper notes that a high β can lose "representations for some factors, particularly those that correspond to only minor changes in pixel space" (Sec. 4.1), and its Figure 6 (right) shows an optimal β for each latent size rather than "the larger the better". Reconstructions get worse steadily as β grows, because less information passes through $\mathbf{z}$. The same figure notes that disentangled representations often come with blurry reconstructions. My two models already show this trade-off: going from $\beta = 1$ to $\beta = 4$ cut the KL from 18.5 to 11.5 nats and the number of informative latents from 6 to 5, while the reconstruction error rose from 39.1 to 63.3 nats. I did not train other values of β, so this answer rests on the paper and on this one pair.

2.10 Implementation Choices

- **Seeds per β: one** (seed 1 for both). One seed can show that this setup produces a mostly disentangled code at $\beta = 4$ and an entangled one at $\beta = 1$, and it gives one concrete comparison between them. It cannot show how often this happens, how large the spread between runs is, or whether the mixed scale latent is typical. The paper's protocol (10 replicas, discard the bottom half) cannot be applied to one run, and the $\pm$ in Table 3 does not include training-seed variability.
- **Training:** batch size 64, 300k steps; minibatches drawn uniformly at random.
- **Columns:** five highest-KL latents above 0.01 nats, ordered by ascending mean σ^2, plus the lowest-KL latent as z_{unused}.
- **Traversal:** seed image an oval at the image center; values $+3, +1.5, 0, -1.5, -3$; Bernoulli mean images.
- **Factor labels:** automatic, from the fraction of each latent's variance explained by each factor. This is not from the paper and is only a guide (it mislabels the rotation latent z_2 in panel A).
- **Disentanglement metric:** $L = 64$ pairs per vector, 10,000 training and 5,000 test vectors, a linear softmax classifier with Adagrad at 10^{-2}, three classifier seeds.

3 Critical Review

3.1 Strengths

The idea is very simple: one extra hyperparameter on a standard VAE. It trains stably, unlike InfoGAN, and needs no knowledge of the factors during training. The qualitative traversals on CelebA, chairs and faces are convincing. The paper also gives the first proper quantitative metric for disentanglement and releases dSprites with known factors. Both of these became standard tools for later work, so the paper had a big influence on how the field evaluates these models.

3.2 Weaknesses

The derivation from the constrained problem is a bit loose. β is called a KKT multiplier, but in practice it is just a fixed hyperparameter. The constraint ϵ is never set or checked. The paper

also does not explain *why* $\beta > 1$ gives disentanglement; it gives intuition and evidence. Later work explained it better through the information bottleneck view [2].

Choosing β needs either the supervised metric (which needs the true factors) or visual inspection. So in practice, the "fully unsupervised" claim depends on some supervision or human judgement when tuning. The paper also says little about how much the results change across random seeds (the appendix just drops the worst 50% of runs). Later work found that the seed can matter as much as β [5]. The comparisons on CelebA, chairs and faces are only qualitative, and the baseline images are reprinted from other papers.

3.3 Potential Errors

- The KL term pushes each *per-sample* posterior $q_\phi(\mathbf{z}|\mathbf{x})$ towards a factorized prior. That does not directly make the *aggregate* posterior $q(\mathbf{z}) = \mathbb{E}_{\mathbf{x}}[q_\phi(\mathbf{z}|\mathbf{x})]$ factorized, which is what independence of the latents actually means. The paper treats the two as the same. Later work splits the KL into a total-correlation term and others to make this precise [4].

- The metric can give a perfect score even when one factor is not captured by any latent. If $K - 1$ factors are disentangled, the classifier can still find the last one by elimination [3]. It also depends on the classifier and on L.

- Figure 7 is from a single trained model per β, so it shows that the effect *can* happen, but not how reliably.

3.4 Comparison with OpenReview Reviews

The reviews were mixed: 6, 7 and 5, plus a public review of 7. The paper was accepted as a poster. The decision note said the area "still seems a little bit conceptually confused", but that the concrete metric and method are original contributions.

Like me, the reviewers liked how simple the method is and saw the metric as a real contribution. Reviewer 1 thought the metric "might come into broader use", which turned out to be true. Most of their criticism was about presentation and evaluation rather than the theory. Two reviewers found Section 3 (the metric) hard to follow. Reviewer 3 wanted quantitative results on real data, not only on dSprites. Reviewer 1 asked why the bottom 50% of scores are thrown away, and the authors said this removes diverged runs. That is related to my point about seeds. Reviewer 1 also noticed that a very high β gives *worse* disentanglement, which seems to go against the "tighter bottleneck is better" story. The authors explained that if β is too high, the capacity drops below the number of factors.

Reviewer 2 was the most negative. They pointed to the missing log-likelihood numbers and asked why we couldn't just use fewer latents instead of a larger β. The authors replied that likelihood is not the point. They also showed that VAEs with fewer latents (sizes 2–10) do not disentangle. Reviewer 2 also asked how β was chosen. The authors said it was picked by eye and that the results were not very sensitive to the seed. This partly answers my concern, but the seed evidence was only in the rebuttal, not in the paper.

None of the reviewers raised my two more technical points: the per-sample vs aggregate posterior issue, and the fact that the metric can be fooled when one factor is missing. Those came up in later papers. So the reviewers focused more on clarity and on evaluation on real data, while my review focuses more on whether the theory actually supports the claim.

References

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